

Transfer Learning for Image Sentiment Classification

A Comparative Study of AlexNet and ResNet



Problem Statement

1

How to optimize visual content in fast-paced digital marketing?

- Images outperform text in capturing attention and conveying emotions
- With high-frequency content posting, it becomes challenging to identify **which images will drive the strongest user engagement**



2

How to Analyze Visual Sentiment in User-Generated Content (UGC)?

- Companies lack tools to **extract sentiment from image-based UGC**
- Existing sentiment analysis research mainly focuses on text, overlooking rich emotional signals in visual UGC

Objectives of the Study

Sentiment Classification

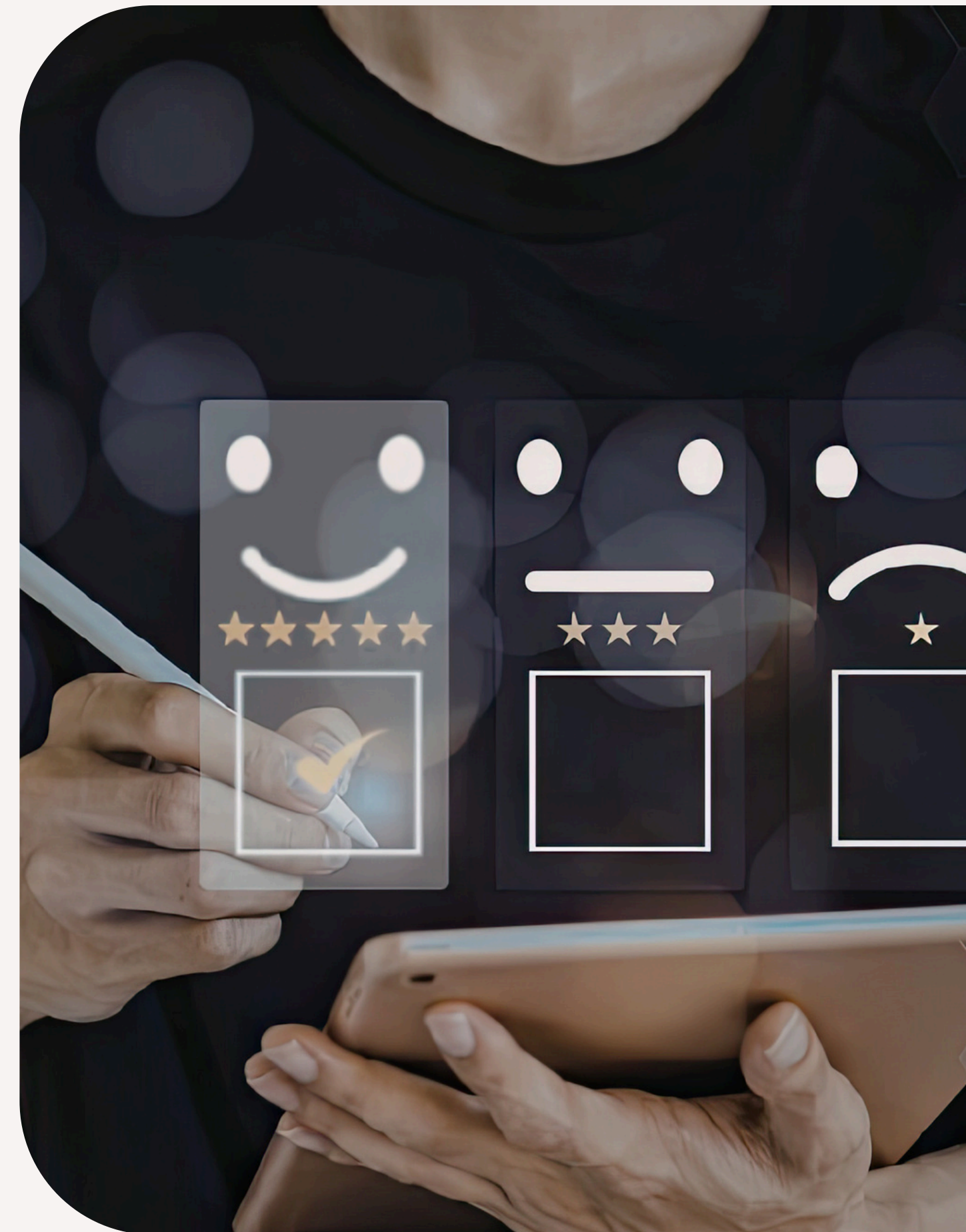
Develop a deep learning model specialized in classify images into four emotional categories: **happiness, sadness, anger, and melancholy**

Engagement Analysis

Use a linear model to examine **how different emotional categories influence Instagram engagement metrics**, identifying which sentiments perform better

Scalable Visual Analysis

Provide **deep learning solution** to replace manual labeling to efficiently process large volumes of **brand-generated and user-generated images**.



Research Question

1

How effectively can deep learning models classify image sentiment across diverse visual content?

- Evaluate the effectiveness of **transfer learning and fine-tuning** in adapting pretrained models (AlexNet and ResNet) for image sentiment classification
- Develop a **specialized deep learning solution** for business-oriented image sentiment analysis

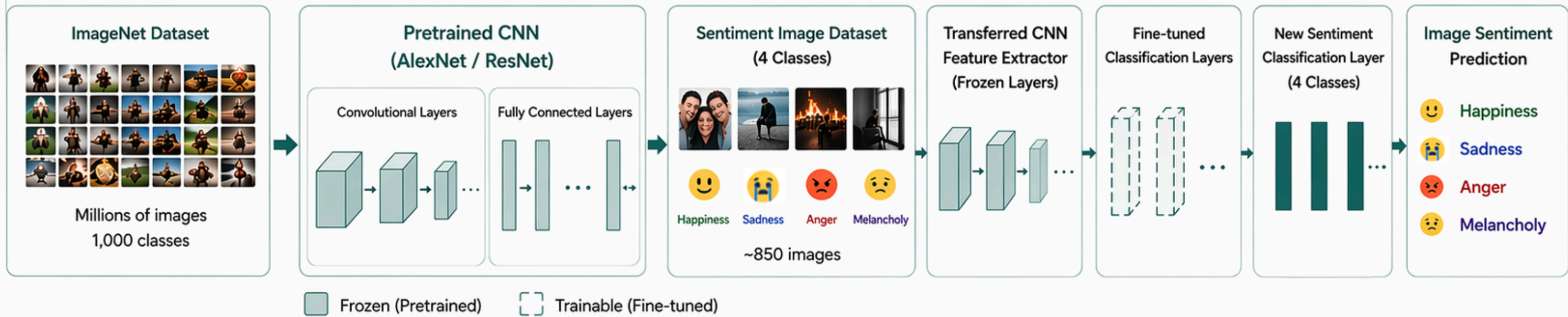
2

What is the relationship between image sentiment and user engagement on social media platforms?

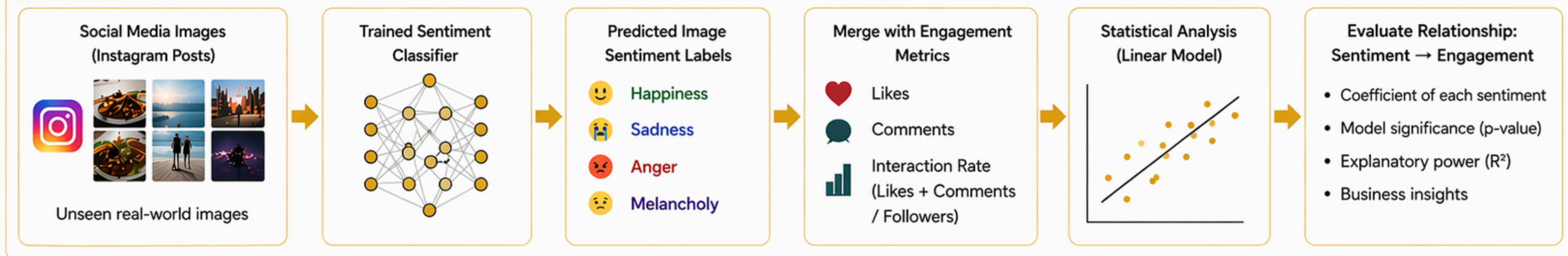
- Evaluate the importance of image sentiment categories in social media post engagement prediction
- Assess the explanatory power of image sentiment in predicting engagement outcomes

Methodology - Research Design

Stage 1 Pretraining on ImageNet & Domain Adaptation (Training on Sentiment Dataset)



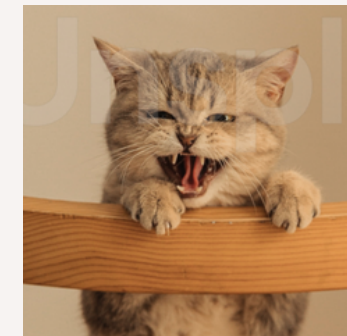
Stage 2 Social Media Application & Engagement Analysis



Methodology - Data Collection

1 Public Dataset: Sentiment from Image

- **Data Source:** Images collected from *Unsplash*, a public photography platform
- **Data Labeling:**
 - **4 Categories:** Happiness, Anger, Sadness, Melancholy
 - **Binary:** Positive vs. Negative
- **Data Size:**
 - **Total:** 850 images
 - **4 Categories:** Happiness (238 images), Anger (240 images), Sadness(186 images), Melancholy (186 images)
 - **Binary:** Positive (238) vs. Negative (612)
- **Characteristics:** the dataset includes **anything capable of conveying emotion**



Angry

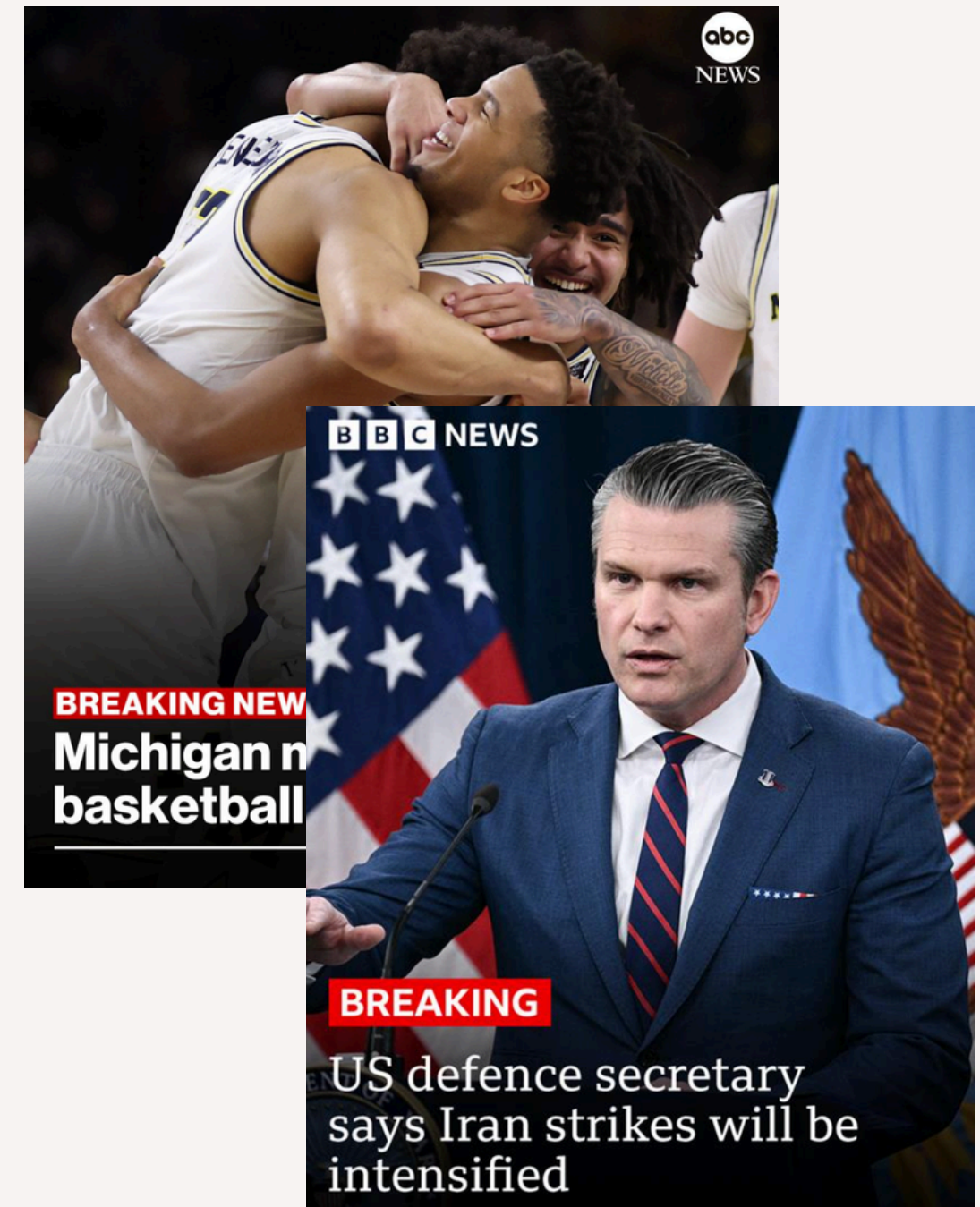


Happy

Methodology - Data Collection

2 Self-collected Dataset: Instagram Posts

- **Data Source:** Real posts from major News media on Instagram, such as the *New York Times*, the *Wall Street Journal*, *BBC News*...
- **Data Structure:**
 - **Engagement Metrics:** Likes, Comments
 - **Content Features:** Multiple image indicator, Post creation timestamp
 - **Account Features:** Follower count, Following count
 - **Calculated Field:** Engagement Rate $(\text{Likes} + \text{Comments}) / \text{Followers}$
- **Data Size:** 100 images



Study 1 - AlexNet

Base Model: torchvision.models.AlexNet (pretrained on ImageNet)
Number of Classes: 4 (Happiness, Sadness, Anger, Melancholy)

Validation Accuracy: 63.24%
Test Accuracy: 63.97%

Training & Test Process

Split dataset

train (70%), validation (15%)
and test (15%)

Data Augmentation

random horizontal flip, 15 degree
random rotation, color jitter

Load pretrained base model (AlexNet)

Evaluate the model

Val Accuracy: 63.24%
Test Accuracy: 63.97%

**Fine-tuning last
convolution block**

- **Learning rate for convolution layers:** 0.0005
- **Learning rate for classifier:** 0.005
- **Epochs:** 30 with early stop
- **Validation accuracy:** 0.6324

Build the transfer learning model

- **Feature Extraction:** Pretrained AlexNet
- **Frozen Layers:** features [0:10] frozen
- **Regularization:** Dropout (0.5)
- **Output:** Linear W/ 4 Class Output
- **Optimizer:** Adam
- **Loss Function:** CrossEntropyLoss
- **Evaluation Metric:** Accuracy

Study 1 Result

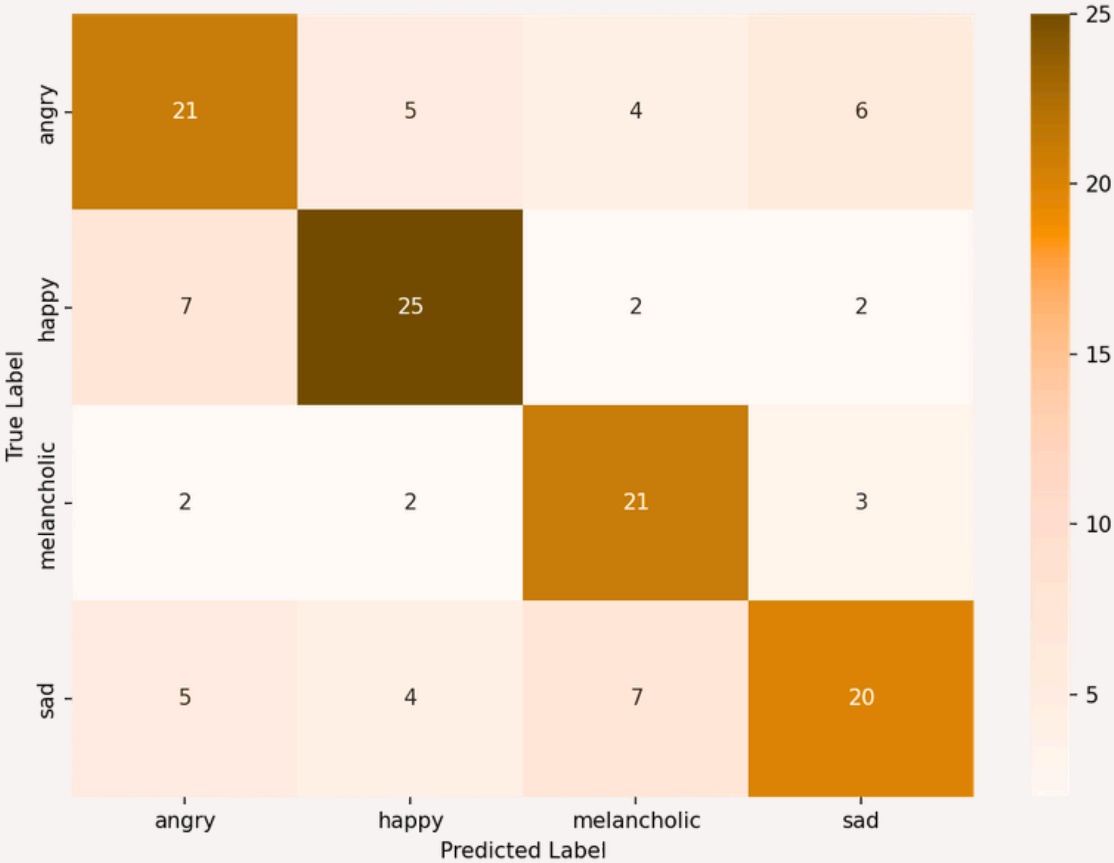
Accuracy 63.97%

Macro F1-score 0.64

Classification Performance

Class	Precision	Recall	F1-score
Happy	0.69	0.69	0.69
Angry	0.6	0.58	0.59
Sad	0.65	0.56	0.6
Melancholic	0.62	0.75	0.68

Confusion Matrix



Findings

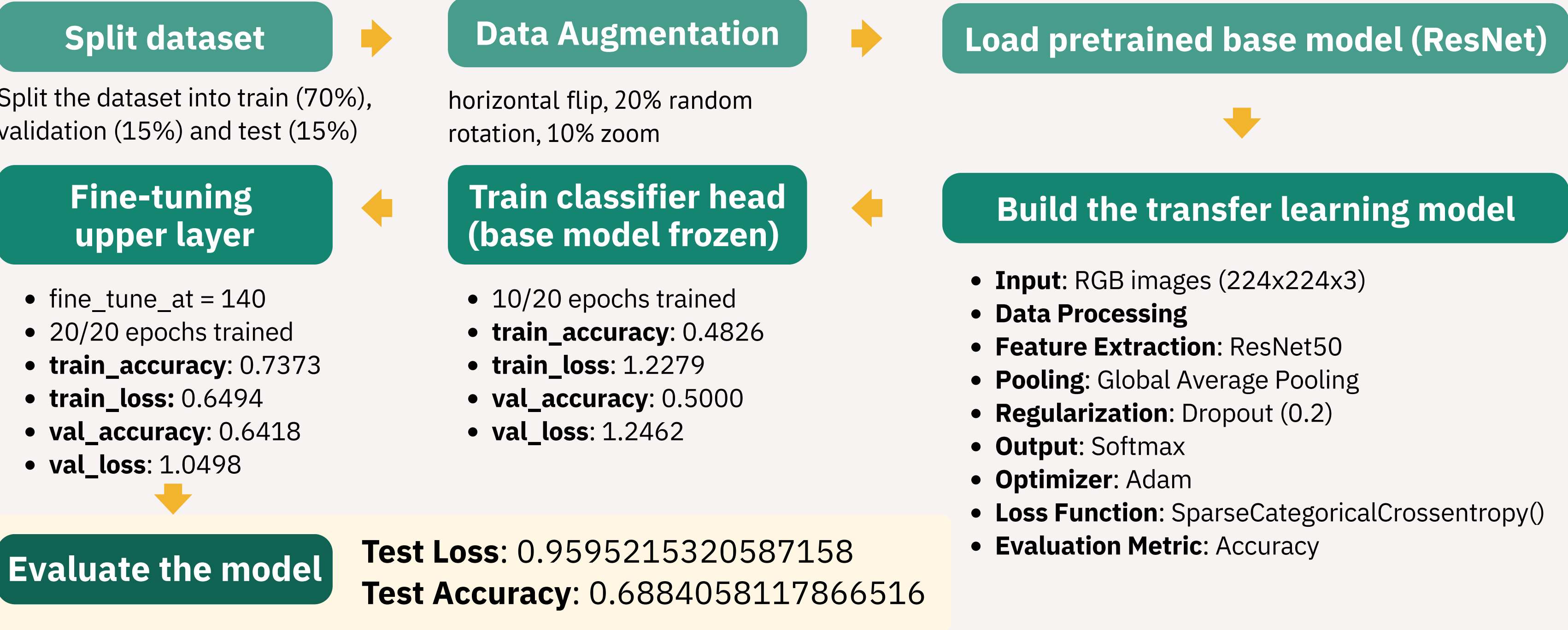
- The AlexNet 4-class model performs better at sentiment analysis than random guessing
- The model performs best on Happiness, while **performing worse on negative emotions** (Angry, Sad, Melancholy), indicating difficulty in distinguishing subtle differences

Study 2 - ResNet

Base Model: ResNet50 (pretrained on ImageNet)
Number of Classes: 4 (Happiness, Sadness, Anger, Melancholy)

Train Accuracy: 73.73%
Test Accuracy: 68.84%

Training & Test Process



Study 2 Result

Accuracy 68.8%

Classification Performance

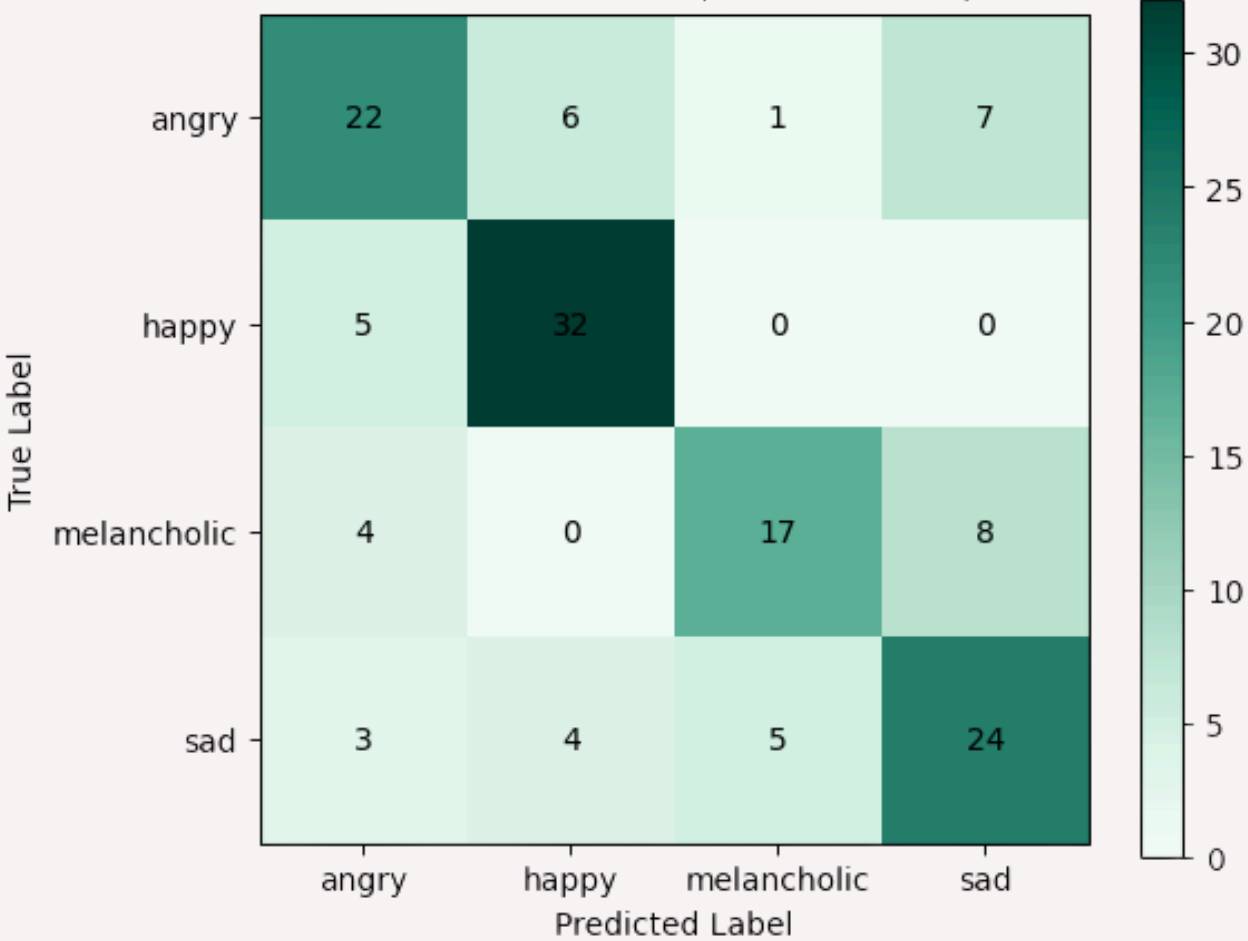
Class	Precision	Recall	F1-score
Happy	0.76	0.86	0.81
Angry	0.65	0.61	0.63
Sad	0.62	0.67	0.64
Melancholy	0.74	0.59	0.65

Findings

- The ResNet50 model significantly improves performance over AlexNet in the 4-class task
- The model performs best on Happiness, while **showing confusion among negative emotions** (Angry, Sad, Melancholy), indicating difficulty in distinguishing subtle differences

Macro F1-score 0.68

Confusion Matrix



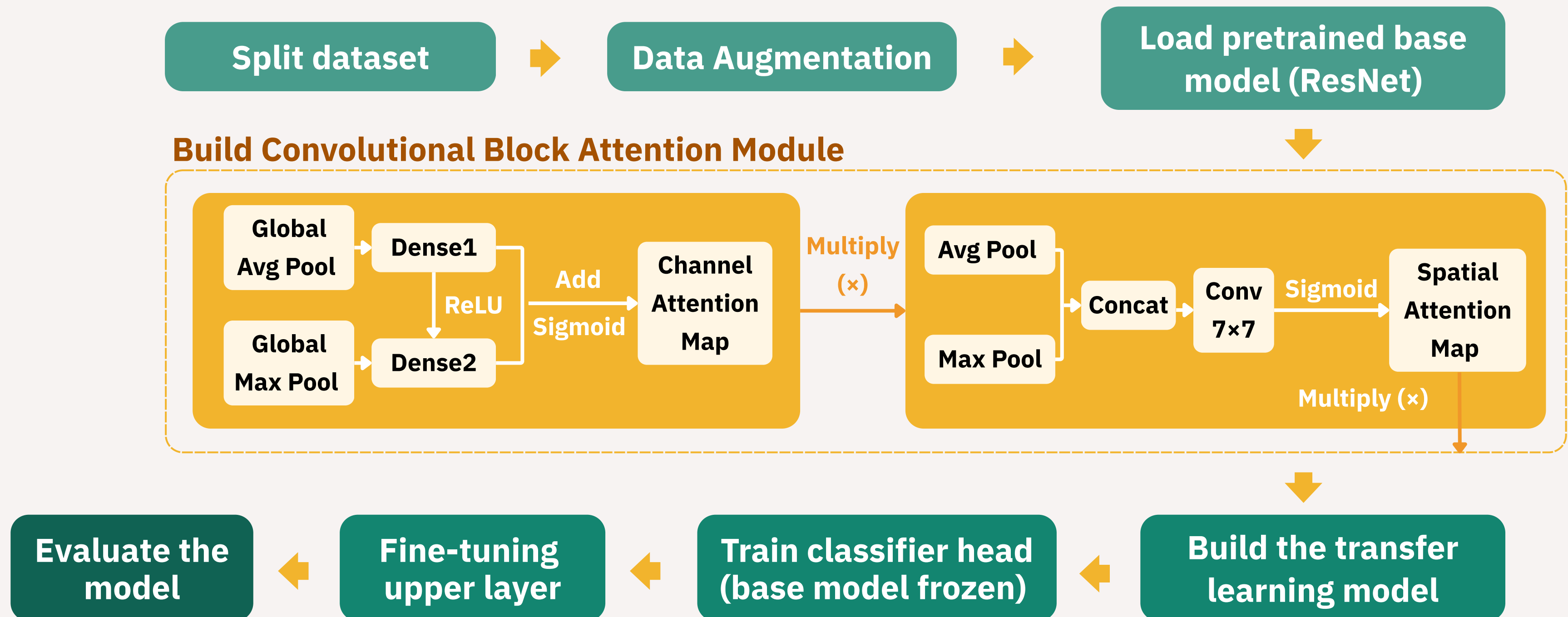
Study 3 - ResNet with Attention Module

Base Model: ResNet50 (pretrained on ImageNet)
Number of Classes: 4 (Happiness, Sadness, Anger, Melancholy)

Train Accuracy: 83.23%
Test Accuracy: **61.59%**

Overfitting!

Training & Test Process



Study 3 - Result (Overfitting)

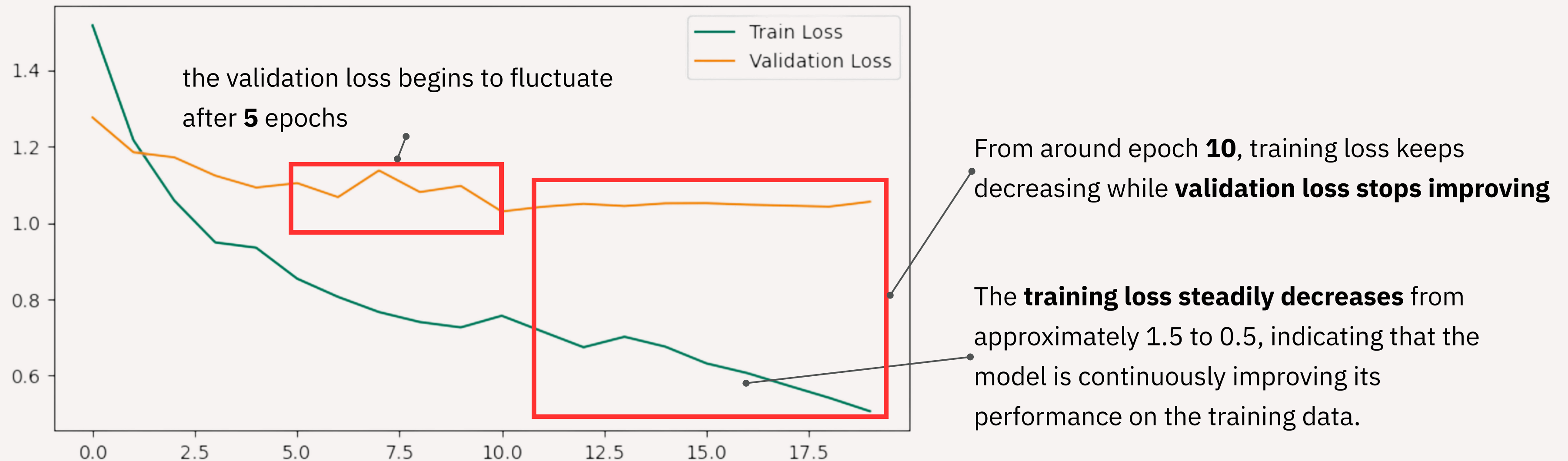
Train Accuracy

83.2%

Test Accuracy

61.6%

Train and Validation Loss Curves



Findings

The model exhibits **overfitting**, as training performance continues to improve while validation performance stop improving, leading to a noticeable drop in test accuracy

Study 4 - AlexNet with Binary task

Base Model: torchvision.models.AlexNet (pretrained on ImageNet)
Number of Classes: 2 (Positive = happy, Negative = angry, melancholic, sad)

Validation Accuracy: 92.65%
Test Accuracy: 89.71%

Training & Test Process

Split dataset

remapped labels, train (70%),
validation (15%) and test (15%)

Data Augmentation

random horizontal flip, 15 degree
random rotation, color jitter

Load pretrained base model (AlexNet)

Build the transfer learning model

- **Feature Extraction:** Pretrained AlexNet
- **Frozen Layers:** features [0:10] frozen
- **Regularization:** Dropout (0.5)
- **Output:** Linear W/ 2 Class Output
- **Optimizer:** Adam
- **Loss Function:** CrossEntropyLoss (weight=[1,3])
- **Evaluation Metric:** Accuracy

Fine-tuning last convolution block

- **Learning rate for convolution layers:** 0.0005
- **Learning rate for classifier:** 0.005
- **Epochs:** 30 with early stop
- **Validation accuracy:** 0.9265

Evaluate the model

Val Accuracy: 92.65%
Test Accuracy: 89.71%

Deep Learning Solution: Study 4 Result

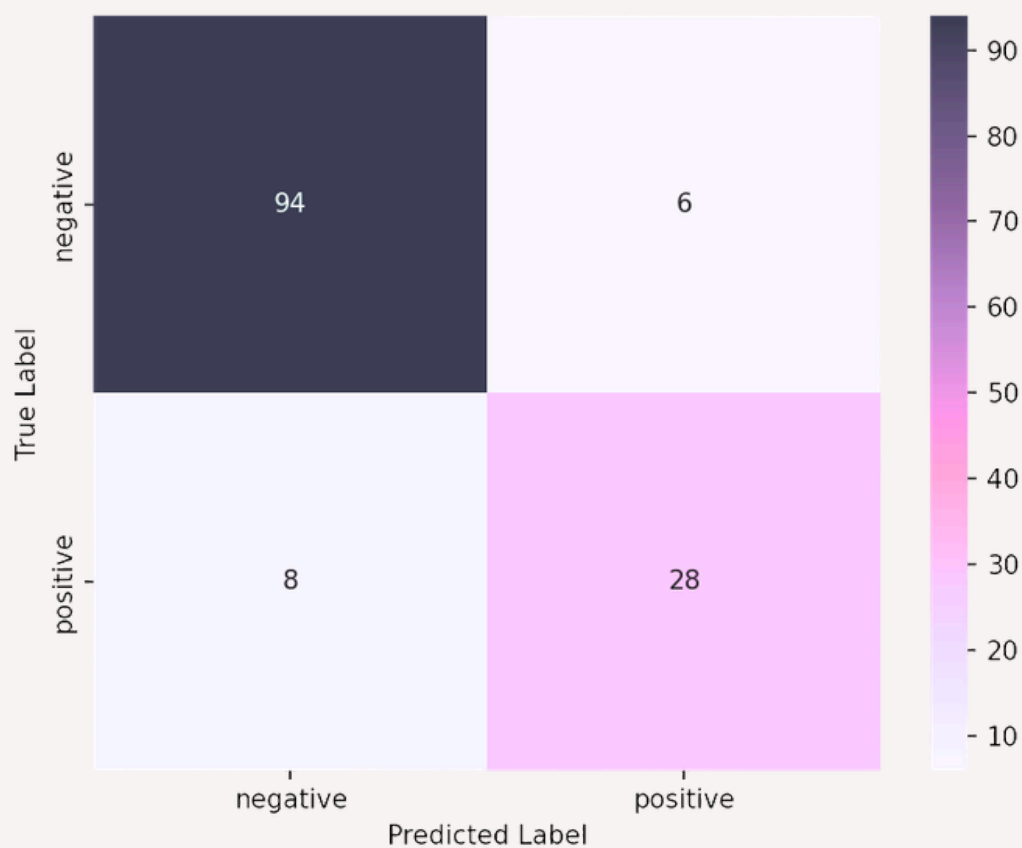
Accuracy 89.71%

Macro F1-score 0.87

Classification Performance

Class	Precision	Recall	F1-score
Positive	0.92	0.94	0.93
Negative	0.82	0.78	0.8

Confusion Matrix



Findings

- The AlexNet binary model has high performance accuracy
- The model still performs best on positive sentiment
- **Major improvements in performance by combining similar sentiments together**

Discussion

Study 1: AlexNet 4-classes

- Test accuracy: 63.97%

Study 2: ResNet 4-classes

- Test accuracy: 68.84%

Study 3: ResNet 4-classes W/ Attention Module

- Test accuracy: 61.59%

Study 4: AlexNet Binary

- Test accuracy: 89.71%
- This model is currently being used for sentiment analysis for our news media dataset

Test Result on News Media Dataset

Predicting News Media Image Sentiment

Good results of sentiment prediction on this unseen data
Only 12 out of the collected images had a confidence score under 80%

Linear Regression For Engagement Rate

Attempted to predict engagement rate, likes, and comments with the sentiment data and other control factors that we included in the data
Every R^2 value was negative, showing that the model was performing worse than simply predicting the mean

Possible Root Causes

Our dataset was not large enough to capture this pattern
Image sentiment alone is not a big enough predictor of engagement
The strength (very positive and very negative vs slightly positive and slightly negative) of the sentiment might play a larger role in predicting engagement than type of sentiment alone